



Central Asian University
in cooperation with “Birlik”
Image processing Internship Program
Major - Computer Science (Senior Level)

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Introduction

Objectives of the Practice

During the internship, students engage in the following tasks based on a competency-based approach to learning:

- 1. Preparation for Solving Software Development and Computational Problems:**
 - Students prepare to tackle software development and computational challenges in a professional environment.
 - They apply theoretical knowledge gained during their studies in computer science and software engineering.
- 2. Familiarization with Software Development and IT Project Management:**
 - Students become acquainted with the organization and methodologies used in software development.
 - They learn about project management practices specific to IT projects.
- 3. Understanding Documentation Procedures:**
 - Students gain insights into the procedures for producing and documenting software solutions.
 - They learn how to create comprehensive documentation for their work.
- 4. Collaboration and Teamwork:**
 - Students develop the ability to work effectively as part of a team.
 - They contribute to software development testing and design efforts alongside their peers.
- 5. Practical Data Preprocessing and Analysis:**
 - Students gain hands-on experience in preprocessing and analyzing data.
 - They apply data science techniques to real-world datasets.
- 6. Preparation of an Internship Report:**
 - Students compile their experiences, learnings, and achievements into a comprehensive internship report.

The Number of Hours for Mastering Work Programs:

4-5 weeks (150 hours) - 5 credits.

Mastering Results of the Practice

Planning and organization of the software development process including professional (PC) and general (GC) competencies:

Code	Name of Learning Outcome
(PC)	To participate in software project planning.
(PC)	To organize project implementation within the framework of the tasks set by the project lead.
(GC)	To understand the essence and social significance of your future profession and show a steady interest in it.
(GC)	To organize your own activities, choose standard methods, and evaluate the effectiveness and quality of performing professional tasks.
(GC)	To make decisions in standard and non-standard situations and take responsibility for them.
(GC)	To search for and use information necessary for the effective implementation of professional tasks and for professional and personal development.
(GC)	To use information and communication technologies in professional activities.
(GC)	To work in a team, communicate effectively with colleagues, management, and clients.
(GC)	To take responsibility for team members' work and the results of completing tasks.
(GC)	To identify tasks for professional and personal development, independently engage in self-education, and plan advanced training in detail.

Thematic Plan of Project-Based Practice

Code and Name of Professional Module	Names of Practice Sections	Quantity (weeks)	Quantity (hours)
Planning and organization of the software development process	Project-based practice for acquiring software development skills and the ability to perform as a team member to develop soft skills	4 to 5	30 to 48 per week
Total		4 to 5	150

Content of the Practice (According to the Profile of the Specialization)

Code of the Professional Module	Planned Educational Result (Practical Experience/Skill)	Types of Work Performed	Content of Works (Detailed Views of Work Performed)
Planning and organization process of software development	To receive practical experience: participation in planning of software development work; participation in the organization of software development work; control and quality fulfillment of software development work.	1. Introductory 1.1. Familiarization with the goals and practice objectives of program practices content of the report. 1.2. Introductory briefing on labor protection and fire security. 1.3 Making workplaces. Learning the structure of enterprises. Rules of the enterprise's internal regulations. 1.4 Studying the structure organization management; official duties of leading professions. 1.5. Familiarization with working documents and coding standards.	- Collecting, processing, and systematizing actual and literary material observations. - Preparing, making up, and submitting a practice report.
Implementation of software development planning	Participation in the collection and use of information about the IT services market.	- Using technical economic and planning indicators in project planning.	- Participating in making documentation for quality management of

	Participation in the compilation of project timelines and schedules using initial documentation.		software products and compiling claims (complaints) related to the quality of materials, products, and finished products.
Ensuring the organization of software development work during the performance of production tasks	-	-	-
Control and evaluation of the quality of project implementation works	-	-	-
Generalization of the material design of the material, a diary (Annex 1), and a report.	-	-	-
Machine Learning Internship	Gain practical experience in developing and implementing machine learning models. Understand and apply different machine learning algorithms. Develop skills in data preprocessing, model training, and evaluation.	- Designing and implementing machine learning models in cloud-based environments - Adapting and developing ML algorithms for business purposes - Supporting development or engineering teams to complete projects	- Evaluating model performance and making improvements
Data Science Internship	Learn to handle, analyze, and visualize data. Develop skills in	- Collecting and cleaning data - Performing	- Working with complex data sets -

	statistical analysis and data mining. Gain experience in using data analysis tools and software.	exploratory data analysis - Creating visualizations to present data insights - Conducting statistical tests and interpreting results	Collaborating in a team environment
Networking Internship	Understand the fundamentals of computer networking. Gain hands-on experience with network setup and troubleshooting. Learn about network security and management.	- Setting up and configuring small networks - Monitoring and troubleshooting network issues - Implementing basic network security measures - Documenting network configurations and changes	- Collaborating with network engineers and administrators
Database Internship	Work with databases, ensuring data integrity, performance, and security. Learn about database management systems (DBMS) and SQL.	- Familiarity with SQL (Structured Query Language) - Understanding of relational databases - Ability to design and optimize database schemas - Knowledge of data normalization and indexing	- Ensuring data integrity and security - Optimizing database performance
Algorithms Internship	Explore algorithm design, analysis, and optimization. Work on solving complex problems efficiently.	- Strong problem-solving skills - Understanding of algorithmic complexity (Big O notation) - Proficiency in at least one programming language - Ability to analyze and optimize algorithms	- Solving algorithmic challenges - Analyzing and optimizing algorithms

Requirements for Educational and Methodological Support of Practice:

Tasks for practice, a diary-report (please see **Annex 1**) on practice, a portfolio for the student, guidelines (instructions) for students for the implementation of all types of work.

Practice Reporting Forms:

- The title page of the report must contain the signature of the head of practice from the enterprise (organization) and be certified by a seal.
- Practice diary (**Annex 1**) - a document reflecting the procedure and timing of the practice as well as instructing students to familiarize themselves with the requirements of labor protection safety and fire safety. The last page of the diary should be signed by the head of practice from enterprises (organizations) and be certified by the seal; the briefing must be certified by the signatures of the student and the head of the practice of the branch organization.
- Joint/group work schedule (plan) for the practice.
- A letter of guarantee from the enterprise confirming the fact that the trainee has completed the internship in a specialized organization indicating the full name of the head of practice from a specialized organization.

Evaluation Criteria:

- Attendance and punctuality (10%)
- Participation and enthusiasm (20%)
- Understanding of development methods (20%)
- Quality of documentation and reports (40%)
- Final presentation and feedback (10%)

Resources and Support Provided by the University:

- Daily report templates (Please see **Annex 1**).
- Access to online learning materials and resources.

Contact Information:

Head of Department:

- Dr. Sajid Khan – s.khan@centralasian.uz

University Coordinators (Project Guides):

- Mr. Pulat Tadjimuratov – p.tadjimuratov@centralasian.uz
- Dr. Adnan Manzoor – a.manzoor@centralasian.uz

- Mr. Khusanbek Gafurov - x.gafurov@centralasian.uz
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Main Sources:

- <https://www.udemy.com/course/how-to-make-a-website-with-wordpress-professional-part-2/>
- <https://www.udemy.com/course/how-to-make-a-professional-website-with-wordpress-2021/>
- <https://www.udemy.com/course/kadence-theme-mastery-course-to-make-websites-with-gutenberg-wordpress/>
- <https://www.udemy.com/course/build-your-own-wordpress-website/>
- <https://www.udemy.com/course/wordpress-39-unveiled-build-your-own-site-fast/>
- <https://www.udemy.com/course/how-to-make-an-ecommerce-business-with-wordpress/>
- <https://www.udemy.com/course/master-figma-in-2024-from-basics-to-pro-level-real-project-ui-ux/>

Other Internet Resources:

- Stack Overflow
 - GitHub
 - Coursera
 - Khan Academy
 - edX
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Annex 1: Design Example - Title Page of Practice Diary

Department of Computer Science

DIARY

Practice for Obtaining Professional Skills and Professional Experience

Week 1 Report

Plans

- Figma - basics
- Figma. Blocks
- Figma. Shapes

Achievements

- I have learnt how blocks are organized in Figma. How to build new ones. There are many features of the blocks. For instance, making columns or rows in order to make easy - designing project. It is really beneficial for designers to take orientation in which position to put elements respectively to each other, so that you can make sure that they are located in the proper destination. It makes our design beautiful.

- Moreover, we can make blocks from other shapes such as RECTANGLES. There are other shapes which could be used. Each figure has its own functionality. The way to make rounded frames for square pictures was the most difficult part in this topic. However, at the end of the day I have learnt how to do this.

Week 2 Report

Plans

- Figma. Images
- Figma. Text
- Figma. list

Achievements

The part of working with images in Figma is not that hard after you learn how to insert them. It is available to crop it. Resizing and changing the position where the photo suits is also some sort of experience. As we mentioned above, in this part blocks with columns and rows come to help you. It was amazing to play with the picture while editing, changing the color and effect.

Text. We can notice that putting texts in Figma is facing most common functions with other editor or designing platforms/programs. Even the way of using almost similar. It is crystal clear that Google docs(word), Google Slides (pptx), Google Drawing (Paint) or Photoshop are really close with Figma from this side of writing and editing the text. You can play with different font styles, size, formatting, coloring and even background coloring...

Lists. This feature makes your work on designing of websites and mobile apps easy and productive instead of doing hard work manually on the text editing and creating lists from scratch.

Week 3 Report

Plans

- WordPress. Paragraphs
- WordPress. Headings
- WordPress. Templates

Achievements:

Paragraphs. Wery similar with Figma's text bar. On the other hand, it is already site not just the design. And this is only for normal paragraphs with small size of letters.

Headings. This is totally different from the previous. Oppositely, it is allowed to change the size of it according to the priority of the title (headings). If you are used HTML then you are familiar with H1, H2, H3, H4, H5, H6 heading sizes.

Templates. The most convenient part of WordPress, if you do not take NOCODE usage of it into account. Google Slides/PowerPoint/Prezi/Canva even the Figma are the best exaples of duplicating and using the template rather than wasting your expensive time on building it from the biggining.

Week 4 Report

Plans

- WordPress. Images and Videos
- WordPress. Gallery vs Slides
- WordPress. Styles and Colors

Achievements:

Images and Videos. The variety of options if working on images absolutely will surprise you. I have never seen such things before: caroozel, gallery, slides and so on. Putting the picture near to the paragraph, rounding the angles or even circling at all... For publishing the video it is necessary to insert the link.

Gallery vs Slides. If you select several images to the gallery it will show all photos at the same time, but when you do it in the slides it is mandatory to press the “NEXT” button to see the next picture. Or we may set in on <<AUTO_PLAY>> mode such that it is going to present all pictures automatically.

Daily Activities Summary

	Date	Module covered	Supervisor's remarks
1st week	3/07/2025	Figma – basics. Udemy	
	4/07/2025	Figma – basics. Practice	

	5/07/2025	Figma. Blocks. Udemy	
	6/07/2025	Figma. Blocks. Practice	
	7/07/2025	Figma. Shapes. Udemy	
	8/07/2025	Figma. Shapes. Practice	
2 nd week	10/07/2025	Figma. Images. Udemy	
	11/07/2025	Figma. Images. Practice	
	12/07/2025	Figma. Text. Udemy	
	13/07/2025	Figma. Text. Practice	
	14/07/2025	Figma. List. Udemy	
	15/07/2025	Figma. List. Practice	

	Date	Module covered	Supervisor's remarks
3rd week	17/07/2025	WordPress. Paragraphs. Udemy	
	18/07/2025	WordPress. Paragraphs. Practice	
	19/07/2025	WordPress. Headings. Udemy	
	20/07/2025	WordPress. Headings. Practice	
	21/07/2025	WordPress. Templates. Udemy	
	22/07/2025	WordPress. Templates. Practice	
4th week	24/07/2025	WordPress. Images and Videos. Udemy	
	25/07/2025	WordPress. Images and Videos. Practice	
	26/07/2025	WordPress. Gallery vs Slides. Udemy	
	27/07/2025	WordPress. Gallery vs Slides. Practice	
	28/07/2025	WordPress. Styles and Colors. Udemy	
	29/07/2025	WordPress. Styles and Colors. Practice	

Supervisor's signature _____